

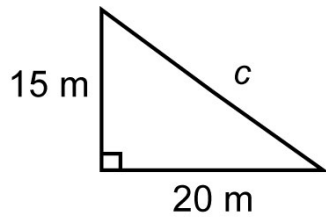
Chapter 8 Review

MPM1D

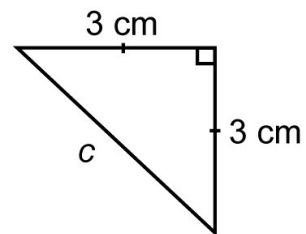
Part A: Pythagorean Theorem

1. Find the length of the unknown side of each triangle:

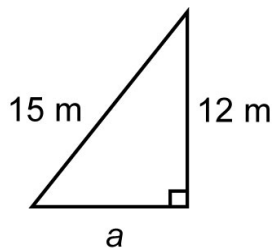
a)



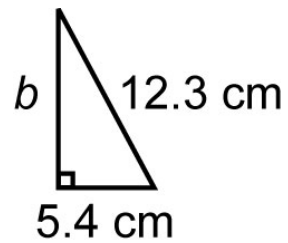
b)



c)

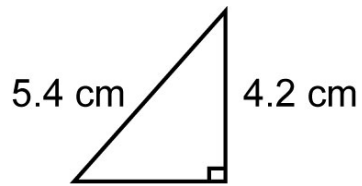


d)

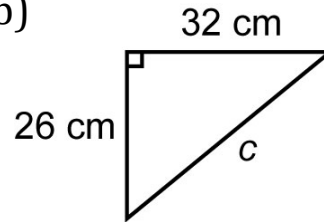


2. Find the area and perimeter of each of the following triangles:

a)



b)



Area = _____

Perimeter = _____

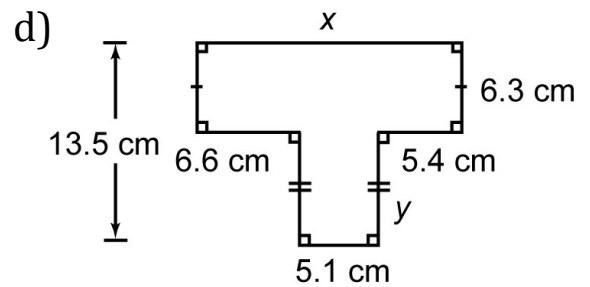
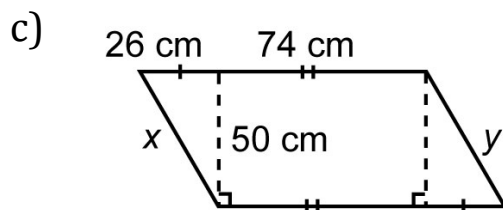
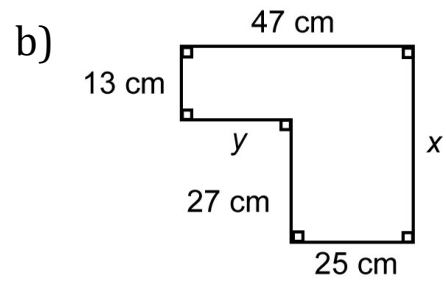
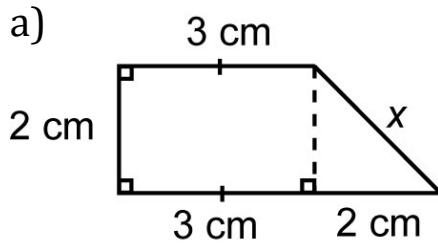
Area = _____

Perimeter = _____

3. A 6-m extension ladder leans against a vertical wall with its base 2 m from the wall. How high up the wall does the top of the ladder reach? Round to the nearest tenth of a meter.

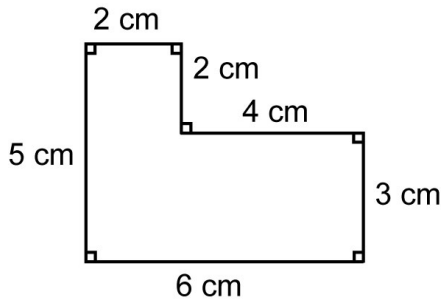
Part B: Perimeter and Area of Composite Figures

4. Find the perimeter of each of the following figures:

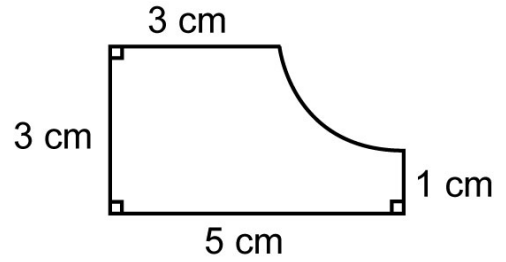


5. Find the area of each of the following figures

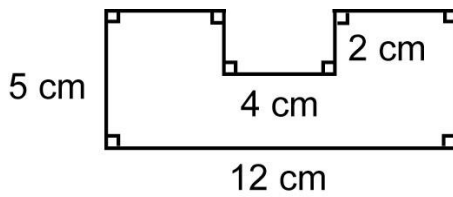
a)



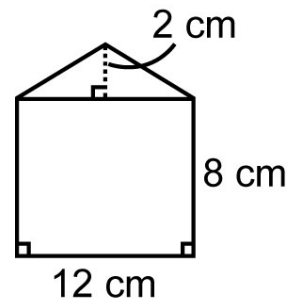
b) *Hint: a quarter circle has been cut out*



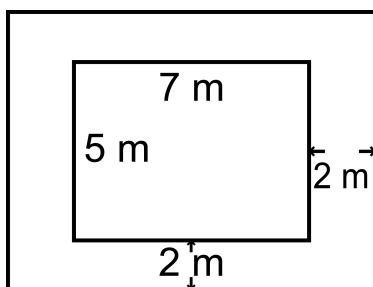
c)



d)



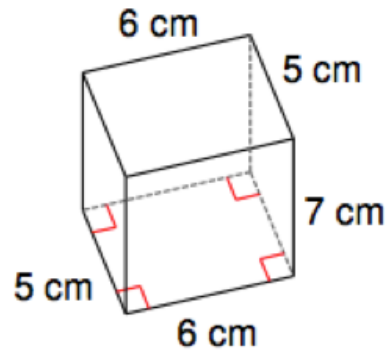
6. A pool is 5 m wide by 7 m long. It is surrounded by a deck 2 m wide. What is the area of the deck?



Part C: Prisms

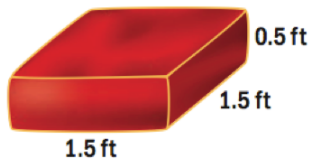
7. For the following rectangular prism:

a) Find the volume:

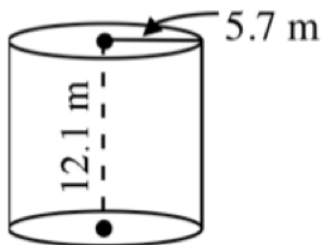


b) Find the surface area:

8. You are making the stadium cushion shown. The foam for the cushion costs \$1.50, and the fabric costs \$.50 per square foot. How much does it cost to make the cushion?



9. Find the volume and surface area of the following cylinder:

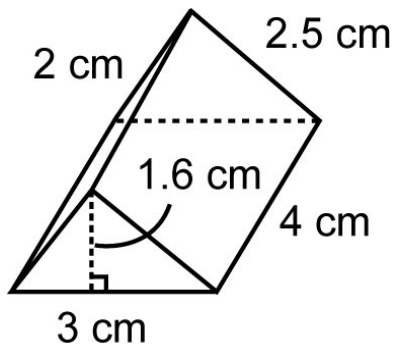


10. A cylindrical tank has a diameter of 2.60 m and a height of 1.65 m.

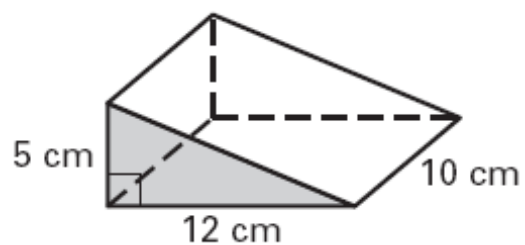
a) Determine the volume of the tank

b) Express the storage capacity in Litres. (Remember $1\text{m}^3 = 1000\text{ L}$)

11. Find the volume and Surface area of the following triangular prism:



12. Here is a picture of a block of cheese Mr. Jensen had for breakfast.

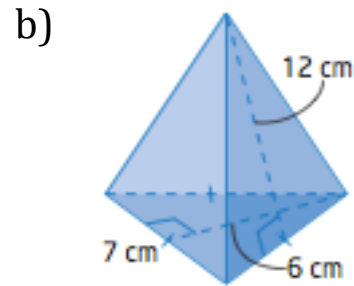
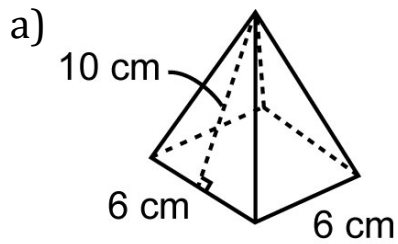


a) Mr. Jensen had to remove the package before eating it. How much packaging did he remove? (Find the surface area)

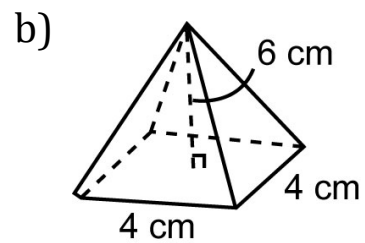
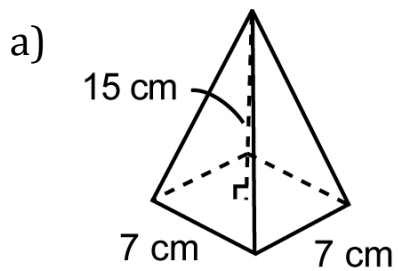
b) How many cubic centimeters of cheese is in Mr. Jensen's stomach right now? (what is the volume of the triangular prism?)

Part D: Pyramids

13. Find the surface area of the following pyramids:

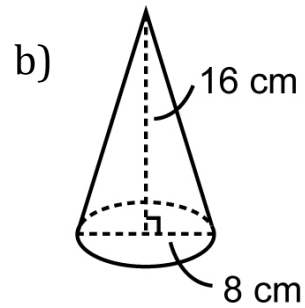
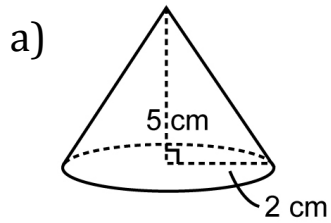


14. Find the volume of the following pyramids

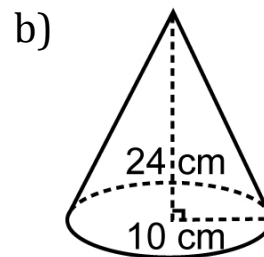
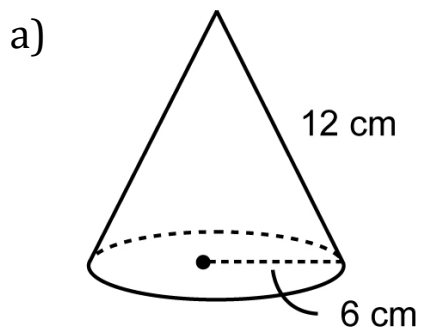


Part E: Cones

15. Find the volume of the following cones:



16. Find the surface area of the following cones:

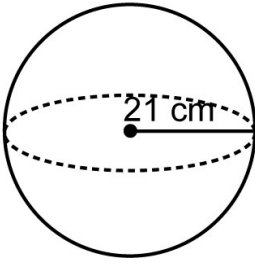


17. A cone has volume 424.1 cm^3 and height 5 cm. What is the base radius of the cone?

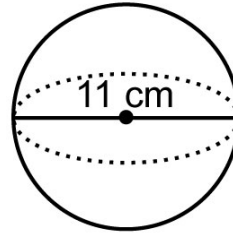
Part F: Spheres

18. Find the volume and surface area of the following spheres

a)



b)



19. Find the radius of a sphere that has Surface Area = 452.4 cm^2

20. A sphere just fits inside a cube with edges 18 cm long.

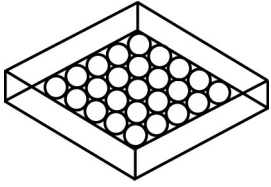
a) Find the volume of the sphere.

b) What is the volume of the cube?

c) How much empty space is there in the cube?

Part G: Application

21. Twenty-five balls, each with diameter 9 cm, are packed in a single layer in a square box.



a) What is the minimum volume of the box?

b) What is the surface area of the box?

c) How much empty space is in the box?

Answers:

- 1) a) 25m b) 4.2cm c) 9m d) 11.1cm
- 2) a) $A=7.14\text{cm}^2$ $P=13\text{cm}$ b) $A=416\text{cm}^2$ $P=99.2\text{cm}$
- 3) 5.7m
- 4) a) 12.8cm b) 174cm c) 312.8cm d) 61.2cm
- 5) a) 22cm^2 b) 11.86cm^2 c) 52cm^2 d) 108cm^2
- 6) 64m^2
- 7) a) 210cm^3 b) 214cm^2
- 8) \$5.25
- 9) $V=1235.1\text{m}^3$ $SA=637.5\text{m}^2$
- 10) a) 8.8m^3 b) 8800L
- 11) $V=9.6\text{cm}^3$ $SA=34.8\text{cm}^2$
- 12) a) 360cm^2 b) 300cm^3
- 13) a) 156cm^2 b) 147cm^2
- 14) a) 245cm^3 b) 32cm^3
- 15) a) 20.9cm^3 b) 268.1cm^3
- 16) a) 339.3cm^2 b) 1131.0cm^2
- 17) 9cm
- 18) a) $V=38792.4\text{cm}^3$ $SA=5541.8\text{cm}^2$ b) $V=696.9\text{cm}^3$ $SA=380.1\text{cm}^2$
- 19) 6cm
- 20) a) 3053.6cm^3 b) 5832cm^3 c) 2778.4cm^3
- 21) a) 18225cm^3 b) 5670cm^2 c) 8682.5cm^3